Intelligent Tutor for Tasks in the Domain of Linear Algebra

Miguel Rodríguez[1[0000−0001−7050−0295]], Eduardo Puraivan[2[0000−0003−2134−8922]], Luis Puebla-Rives[1[2222−−3333−4444−5555]], Juan Pablo Ruiz-Rodríguez[3[0000−0002−2112−4622]], Diego Saavedra-Lizana[1[0009−0003−6773−1473]], Mauricio Gamboa[4[0000−0002−0315−4334]], Connie Cofré-Morales[1,2[0009−0000−1442−1218]], Karina Huencho-Iturra[2[0009−0009−7043−2929]], and Pablo Gregori-Huerta[5[0000−0002−1306−341X]]

[1] Universidad de Playa Ancha, Chile
{mrodriguez, connie.cofre}@upla.cl
[2] Universidad Viña del Mar
epuraivan@uvm.cl, Karina.huencho.iturra@alumno.uvm.cl
[3] Universidad de Chile, Chile
juan.ruiz@ing.uchile.cl
[4] Universidad de Concepción, Chile
maurigamboa@udec.cl
[5] Universitat Jaume I de Castellón, España
gregori@uji.es

Abstract. Difficulties in learning linear algebra are well-documented and arise from epistemological, didactic, and instructional factors. This study presents the design of an affective intelligent tutor focused on the mathematical object known as systems of linear equations (SLE). The tutor provides adaptive feedback and was developed using an iterative methodology that combines design, internal validation, and continuous improvement. It integrates several components: a didactic knowledge module, a model of ideal student behavior, a pedagogical tutoring module, a graphical interface, and an affective module based on computer vision. This affective component employs the DeepFace library, integrated with OpenCV and PyQt6, to detect emotions in real time through facial expressions. Emotional inputs are processed to compute a stress index, which triggers adaptive, affective feedback during task execution. The current version includes one task designed to help students coordinate the seven different representations of a 3×2 SLE system, thereby enhancing their understanding of the solution set. To date, expert and functional validation has been conducted. The tutor shows potential as a support tool for autonomous learning and as a training resource for future mathematics teachers. Preliminary testing with experts revealed challenges such as redundant emotion detection and visual overlap of interface elements. These findings have informed subsequent redesign efforts. Although still in the prototype phase, the system shows promise in supporting representational coordination in tasks involving SLE.

Keywords: linear algebra · systems of linear equations · affective computing · intelligent tutor · feedback

1 Introduction

The Babylonian, Greek, and Chinese civilizations addressed practical problems in trade and land surveying using using SLE [1, 9], leading to methods like the Fangcheng shu—structurally similar to the Gauss-Jordan method [9]. In the 17th and 18th centuries, SLE were applied to curve fitting, and by the end of that period, Euler studied them as autonomous mathematical objects, showing that the nature of their solutions depends on the relationships among the equations [6]. This formalization process faced challenges: the dominance of algebraic methods often obscured the geometric aspects of solving SLE. Still, it marked the emergence of the concept of linear dependence, later formalized and extended beyond the context of SLE [6, 8].

On the other hand, from an educational perspective, considering that the processes of abstraction, generalization, and axiomatization that took place in mathematics to shape the fundamental concepts leading to what we now know as Linear Algebra (LA) took centuries and faced many difficulties. This inevitably creates difficulties for students learning these topics in university, preventing them from fully understanding the meaning and scope of this new knowledge [6]. In particular, this has been evidenced in research related to the understanding of SLE, and more specifically 3×2 systems, as this requires a complex articulation between different mathematical representations—algebraic, graphical, and structural—that students do not always manage to establish fluently [14, 15].

In relation to the above, research in mathematics education has shown that students' difficulties in this domain of mathematics do not stem solely from procedural shortcomings, but also from epistemological obstacles related to the very nature of the mathematical object [6]. Added to this are difficulties inherent in traditional teaching, which often prioritizes algorithmic methods over conceptual understanding and geometric visualization [12, 13].

In response to this scenario, the development of an intelligent tutor with affective feedback is proposed. Its purpose is to guide the student in the active exploration of the seven possible geometric configurations of a 3×2 SLE, in order to support the cognitive construction of the solution set of this mathematical object [15].

The design of this tool is based on a dual inspiration: on the one hand, the classical theoretical frameworks of ITS, and on the other, didactic research addressing the learning difficulties associated with SLE, from perspectives such as APOS Theory [2] (Action, Process, Object, Schema) and modes of mathematical thinking. This article presents the design foundations of the tutor, its architecture, the tasks it proposes to students, and the preliminary results obtained in the early stages of development.

Despite the historical evolution of linear algebra and the depth of its conceptual structure, few educational technologies explicitly integrate affective and

didactic components to address learning obstacles in SLE. This project responds to this gap by proposing an intelligent tutor that not only supports problem solving but also incorporates pedagogical principles and real-time emotional feedback to foster deeper conceptual understanding. As highlighted by [18], the integration of affective computing in ITS enables these tools to detect and respond to learners' emotional states, significantly enhancing engagement and conceptual comprehension in cognitively demanding tasks. This approach aligns with recent innovations in educational technology, where emotion-aware ITS are being designed to personalize instruction based on learners' affective and cognitive profiles, offering a more holistic and responsive learning environment. Furthermore, [19] demonstrated that combining peer instruction with seminar strategies and didactic engineering in linear algebra courses fosters active learning, enhances motivation, and improves conceptual mastery, suggesting that integrating such pedagogical strategies into intelligent tutors could further support students in overcoming learning obstacles in SLE.

2 Related work

2.1 Intelligent Tutoring Systems (ITS)

ITS represent a convergence of artificial intelligence, learning theory, and educational technologies aimed at providing personalized teaching experiences. Unlike traditional computer-assisted instruction systems, ITS explicitly integrate structured domain knowledge, a student model, pedagogical strategies, and an adaptive interface [14]. This modular architecture enables the diagnosis of errors, the generation of meaningful feedback, and the dynamic adjustment of instruction according to the learner's cognitive and emotional needs.

According to [4, 10], an ITS is structured around four core components: (1) an expert knowledge module, (2) a student model, (3) a tutoring module, and (4) a user interface. These modules work together to provide an educational experience capable of diagnosing errors, proposing corrective strategies, and adapting to the learner's individual needs. Our tutor is based on this architecture and additionally incorporates an affective module, which responds to the student's emotional state through motivational feedback.

Recent developments in the field of ITS highlight the increasing integration of affective computing to enhance learning personalization and engagement. In [18] it is emphasize that systems capable of detecting and responding to students' emotional states can significantly improve motivation and cognitive performance. In mathematics education, [17] provides a comprehensive review of ITS implementations using the SAMR model and notes that most systems remain at the augmentation stage, with few achieving deep transformations in pedagogical practices. Affective feedback mechanisms, as examined by [20], offer new pathways to emotional self-regulation in learners, especially in tasks with high epistemic demand. Furthermore, a recent review by [7] underlines the importance of designing affect-aware educational technologies that go beyond task completion, focusing instead on conceptual understanding and student autonomy. In

comparison to widely used systems such as ALEKS and ASSISTments, which primarily focus on adaptive sequencing and procedural accuracy, our proposal incorporates emotional monitoring, conceptual scaffolding, and active manipulation of mathematical objects. This positions our tutor within a new generation of ITS that combine domain-specific didactic knowledge, real-time affective feedback, and exploratory environments to support the construction of meaning in advanced mathematical tasks such as systems of SLE.

2.2 Affective Tutoring and Emotional Feedback

Recent studies, as synthesized by [4], have highlighted the crucial role that emotions play in learning processes, particularly in contexts mediated by intelligent technologies. The integration of an affective module in an ITS enables the detection of students' emotional states—such as frustration, confusion, or lack of motivation—and the activation of adaptive responses aimed at sustaining engagement and promoting both emotional and cognitive self-regulation. In this study, emotional classification is based on the dimensions of valence and arousal, allowing for the selection of motivational phrases that are contextually relevant and tailored to the user's affective profile.

Affective feedback has proven especially effective in learning environments where cognitive challenge can lead to blockage or disengagement. Its implementation in this tutor seeks not only to support students in solving mathematical tasks, but also to provide emotional scaffolding at critical moments of the learning process.

2.3 Epistemological Difficulties in Learning SLE

Learning SLE, particularly in the 3×2 case, presents challenges that go beyond procedural aspects. Research based on the APOS Theory has shown that many students fail to construct a coherent schema of the "system" object or to connect algebraic solutions with their geometric representations [15].

Likewise, from the perspective of modes of thinking (synthetic-geometric, analytic-arithmetic, and analytic-structural), it has been shown that students tend to operate within a single mode without being able to shift between them, which limits their overall understanding. These difficulties, documented in both secondary and university students, justify the choice of a task that requires coordinating representations and reflecting on the structure of the SLE [12].

The tutor presented in this work seeks to address these issues not only through technological support, but also through a didactic proposal that is mindful of the epistemological complexity of the mathematical object in question.

3 Methodology

The proposed model consists of three interconnected modules, whose interaction simulates the dynamics between a student and a teacher in an educational

setting. This architecture, based on a generic model proposed by [10], aims to computationally represent the relationships established during the resolution of a test situation, as shown in Figure 1.

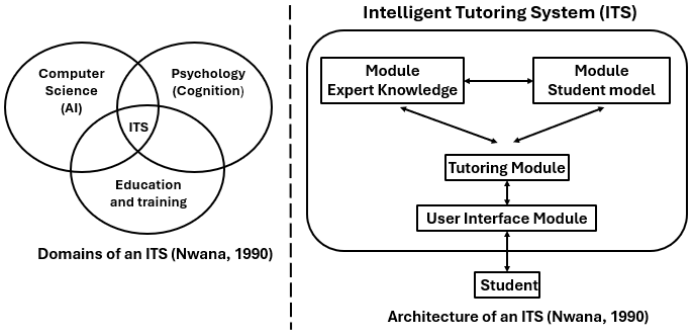

Fig. 1. Structural Elements of the Intelligent Tutor.

The core of the system is a playground-type interactive environment, in which the student can engage with a series of problems that allow for a hands-on approach to a specific content area. In this work, that environment consists of a visual representation of a Cartesian plane, where the student can manipulate lines to solve exercises related to collinearity. This module incorporates an emotional sensing system based on computer vision, which detects the student's facial expressions to infer their emotional state during problem-solving, as shown in Figure 2.

Additionally, the system integrates an intelligent tutor module, which processes the stimuli collected from the student and provides adaptive feedback. This tutor is structured into three distinct submodules:

1. **Knowledge base**: Contains the theory associated with the teaching content, including a searchable repository of frequently asked questions, solved problems, and conceptual explanations. It can be based on an encyclopedia-style model or be accessible via an LLM to facilitate interaction.
2. **Student model**: Estimates the student's level of conceptual mastery, problem-solving strategy, and emotional response based on the context. It can learn from the student's interaction with the interface via reinforcement learning to enable adversarial training with the tutor module.
3. **Tutor module**: Generates instructional stimuli and specific guidance based on the estimated student profile. It can learn from interaction with a real tutor via reinforcement learning, although it would initially be based on a rule-based logic.

Both the student model and the tutor module can be trained using data generated from the interaction between the user and the environment, as well as through teacher interventions.

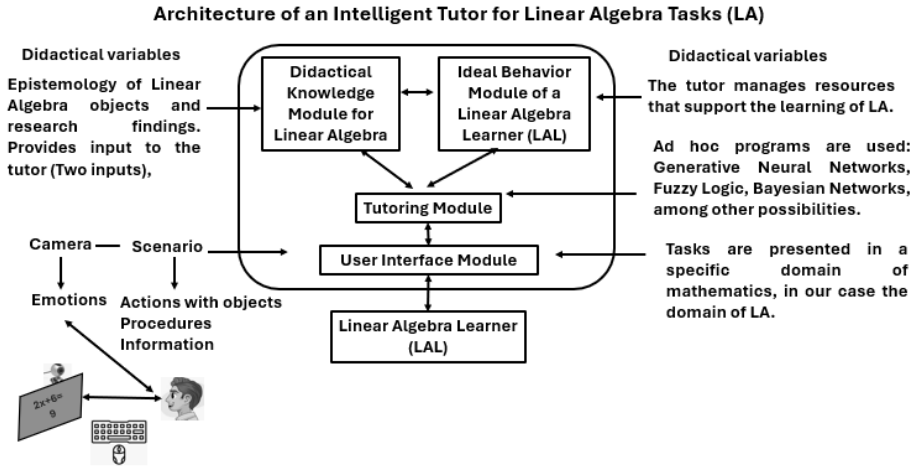

Fig. 2. Intelligent Tutor Architecture.

During the experience, the student interacts with a set of prototypical problems in the playground, facing varying degrees of familiarity with the tasks presented. This process leads to particular emotional responses (e.g., stress or frustration), variable cognitive strategies, and possible queries directed to the tutor, all of which are collected for the system's adaptive feedback.

3.1 General Design Approach

The intelligent tutor developed in this work was designed based on the classic principles of ITS, also integrating an emotional module that provides affective feedback during the resolution of a complex mathematical task. The proposal was conceived as a support tool for students who face difficulties in the structural understanding of 3×2 linear equation systems.

The design is based on an iterative methodology, coordinated between a team of researchers in mathematics education and a software developer. The first functional version of the system has been evaluated through internal review sessions, focusing on pedagogical coherence, the quality of the interface, and the tutor's behavior with different user profiles.

The first iteration was evaluated by a team of mathematics education researchers through internal expert reviews. These sessions provided insights into graphical usability, emotional feedback accuracy, and pedagogical coherence. No student data was collected at this stage.

3.2 System Architecture

The affective module of the tutor is based on the premise that emotions significantly influence cognitive processes and, therefore, learning [5, 11]. With the goal

of emotionally supporting the student during the resolution of complex mathematical tasks, a real-time facial recognition system was integrated to infer the user's emotional state and activate personalized affective feedback. This system was implemented using the DeepFace library, a deep learning-based facial analysis tool [16], complemented with OpenCV for continuous image capture from the device's camera. Every two seconds, the system captures an image of the student's face and performs an inference that returns the predominant emotion. The emotions considered include: happy, sad, fear, angry, neutral, surprise, and disgust.

Following the affective valence model described in [3], which classifies emotions based on their positive or negative charge, a subjective scoring scale was defined for each emotion recognized by the system. This scale allows the student's recent emotional state to be quantitatively represented through a normalized stress index. Negative emotions such as *sad*, *fear*, and *angry* are assigned a score of +1, while positive emotions like *happy* receive a score of -1. Ambiguous emotions such as *surprise* and *disgust* are assigned a score of 0.5, and *neutral* is considered the midpoint with a value of 0. This index is calculated using the following expression:

$$\text{Stress Index} = \frac{1}{2}\left(\sum_{i=1}^{n} score(\text{emotion}_i)\right)$$

Where n is the number of emotions recorded during the analysis interval, and $\text{score}(\text{emotion}_i)$ represents the score assigned to the emotion at instant i. The formula normalizes the weighted average to the interval $[0, 1]$, where values close to 1 reflect high-stress states, and values near 0 indicate emotionally positive or regulated states. Table 1 presents a summary of the emotional scoring system used.

Table 1. Scores assigned to each emotion.

Emotion	Score assigned
happy	-1
sad, fear, angry	$+1$
neutral	0
surprise, disgust	0.5

Based on this index, the system deploys three types of affective interventions, according to predefined temporal and emotional criteria:

- **High stress**: If the index exceeds 0.6 and at least 30 seconds have passed since the last alert, a message is displayed acknowledging the emotional load and offering explicit support.
- **Distraction or moderate stress**: If the value falls between 0.4 and 0.6, and at least 5 minutes have passed, a notification is triggered suggesting a break or redirection.
- **Sustained positive emotional state**: If the index remains below 0.4 for at least 10 minutes, the system issues a positive reinforcement message.

The integrated timers ensure that affective interventions are neither intrusive nor redundant, respecting the student's pace and autonomy. This emotional feedback mechanism, far from being a merely decorative feature, represents a critical functional layer for the tutor's adaptability, as it modulates the user experience based on dynamic affective variables. In doing so, the system moves toward a truly learner-sensitive tutoring model, integrating cognition and emotion within a unified pedagogical flow. The system consists of five main modules:

- **Didactic knowledge module**: Incorporates the theoretical and didactic foundations of the addressed topic, based on previous research.
- **Ideal student behavior model**: Contains the expected actions and responses for each of the proposed tasks.
- **Tutoring module**: Manages system interventions in response to errors, stagnation, or task deviations.
- **Affective module**: Detects emotions through facial recognition and activates relevant emotional phrases.
- **Graphical interface**: Allows the student to construct and manipulate geometric representations of systems of equations.

The architecture follows the classic framework of [10], enriched with features typical of mixed tutors (shared control between the system and the student).

Although a formal accuracy validation of the emotion recognition system has not yet been conducted, controlled testing with the DeepFace + OpenCV integration confirmed reliable basic performance. Future empirical phases will incorporate user consent protocols to address privacy concerns.

3.3 Description of the Mathematical Task

The proposed task consists of graphically modeling the seven possible configurations of a 3×2 SLE by representing three lines in the plane. The key point is for the student to understand what a solution is and what can happen with the solution set of a $3x2$ linear system.

The interface allows the student to draw, label, move, and analyze the lines, as well as identify common points and record conclusions. This activity aims to promote the coordination of algebraic and geometric representations, in addition to stimulating transitions in the way of visualizing the objects.

4 Results

4.1 Preliminary Observations of the Prototype

The first functional version of the tutor was subjected to a series of internal testing sessions, focused on analyzing technical performance, pedagogical coherence, and user experience quality. These tests allowed the identification of strengths as well as areas for improvement. Among the most relevant observations are the following:

- **Early presentation of the solution**: In the introduction of the task, the system shows an image with all possible configurations of the 3×2 system, which contradicts the proposed exploratory approach. It has been decided to remove this image in future versions.
- **Rigid graphical interface**: The equations of the lines overlap and cannot be relocated, making it difficult to clearly visualize the student's work. It is proposed to separate the visualization of the lines and their equations and allow manual editing.
- **Repetition of emotional states**: The system constantly detects and reports the same emotional state (for example, "neutral"), leading to irrelevant emotional interventions. This suggests the need to refine the recognition model and enrich the available phrase base.
- **Lack of adaptive help**: Currently, the tutor does not offer specific support when the student gets stuck or makes errors. It is proposed to implement a decision tree-based system to generate contextual interventions based on the student's actions.

During testing, several recurring behaviors were noted. For instance, the system repeatedly detected the same emotion ("neutral") without adjusting, and students found it difficult to distinguish overlapping lines. These findings informed decisions to improve emotional sensitivity and graphical clarity.

4.2 Adjustments in Development

Based on these observations, several lines of improvement have been defined, which are currently in the implementation stage:

- Redesign of the interface with more precise graphical tools and dynamic labels (L1, L2, L3).
- Expansion of the emotional recognition system with higher levels of sensitivity and customization.
- Incorporation of a strategic help module based on interaction patterns and common errors.
- Optimization of traceability logging for subsequent analysis of the resolution process.

These advances will allow for the evaluation, in a second stage, of the tutor's instructional effectiveness with real students, both in terms of mathematical performance and emotional experience during the task.

5 Conclusions

The intelligent tutor presented in this work constitutes a didactic and technological proposal aimed at supporting the understanding of 3×2 SLE, through the integration of graphical tools, an intelligent architecture, and an adaptive

affective component. Its design is grounded in a review of the epistemological difficulties commonly faced by students in this domain, as well as in well-established principles of ITS.

Preliminary results suggest that the system may provide a suitable foundation for fostering transitions between representations and modes of thinking. However, several key areas for improvement have been identified, particularly regarding the graphical interface, emotion recognition, and the timely activation of contextual support.

Looking ahead, empirical validation of the tutor is planned in real educational settings, which will enable the assessment of its potential impact on conceptual understanding, motivation, and learning autonomy. Additionally, the architecture is expected to be adapted to address other complex topics in Linear Algebra, maintaining the focus on the articulation between representations and the consideration of emotional aspects in mathematical learning.

The intelligent tutor presented here represents a didactic-technological innovation that integrates graphical modeling, real-time emotion detection, and conceptual feedback grounded in mathematics education research. Future empirical validation will evaluate its effectiveness in enhancing students' understanding of SLEs and regulating their emotional responses. Furthermore, the architecture is designed to be scalable and adaptable to other topics in linear algebra, such as vector spaces and linear transformations, always maintaining its emphasis on the coordination of representations and emotional engagement.

Acknowledgment

This work is supported by: the Call for Technical Support for R+D+i and Creation Functional Units, UPLA, 2024/2025 Edition, and by the Fondecyt Initiation Project No. 11190152.

References

1. Andrews-Larson, C.: Roots of linear algebra: An historical exploration of linear systems. PRIMUS **25**(6), 507–528 (Apr 2015). https://doi.org/10.1080/10511970.2015.1027975
2. Arnon, I., Cottrill, J., Dubinsky, E., Oktaç, A., Roa Fuentes, S., Trigueros, M., Weller, K.: APOS Theory: A Framework for Research and Curriculum Development in Mathematics Education. Springer New York (2014). https://doi.org/10.1007/978-1-4614-7966-6
3. Calvo, R.A., D'Mello, S.K.: Affect detection: An interdisciplinary review of models, methods, and their applications. IEEE Transactions on Affective Computing **1**(1), 18–37 (2010). https://doi.org/10.1109/T-AFFC.2010.1
4. Carbonell Bernal, N., Hernández Prados, M.: Impacto de los sistemas de tutoría inteligente. una revisión sistemática. Edutec. Revista Electrónica de Tecnología Educativa **89**, 121–143 (Sep 2024). https://doi.org/10.21556/edutec.2024.89.3025
5. D'Mello, S., Graesser, A.: Autotutor and affective learning. In: Calvo, R.A., D'Mello, S.K. (eds.) New Perspectives on Affect and Learning Technologies, pp. 113–126. Springer, New York, NY (2012)

6. Dorier, J.L. (ed.): On the Teaching of Linear Algebra, Mathematics Education Library, vol. 23. Kluwer Academic Publishers (2002). https://doi.org/10.1007/0-306-47224-4
7. Fernández-Herrero, J.: Evaluating recent advances in affective intelligent tutoring systems: A scoping review of educational impacts and future prospects. Education Sciences **14**(8), 839 (Aug 2024). https://doi.org/10.3390/educsci14080839, http://dx.doi.org/10.3390/educsci14080839
8. Kleiner, I.: A History of Abstract Algebra. Birkhäuser Boston (2007). https://doi.org/10.1007/978-0-8176-4685-1
9. Martzloff, J.C.: A History of Chinese Mathematics. Springer (2006)
10. Nwana, H.: Intelligent tutoring systems: an overview. Artificial Intelligence Review **4**(4) (1990). https://doi.org/10.1007/bf00168958
11. Picard, R.W.: Affective Computing. MIT Press, Cambridge, MA (1997)
12. Rodríguez, M., Mena-Lorca, A., Gregori, P., Vásquez, P., del Valle, M., Parraguez, M.: Comprensión del conjunto solución de un sistema de ecuaciones lineales de dos incógnitas: un estudio de casos. Educación Matemática **34**(3), 163–193 (Dec 2022). https://doi.org/10.24844/em3403.06
13. Rodríguez, M.A., Gamboa, M.A., Aros, V.H., Vergara, A.S.: La construcción cognitiva del r3 espacio vectorial: un estudio de caso. Formación universitaria **17**(4), 55–70 (Aug 2024). https://doi.org/10.4067/s0718-50062024000400055
14. Rodríguez Chávez, M.H.: Sistemas de tutoría inteligente y su aplicación en la educación superior. RIDE Revista Iberoamericana para la Investigación y el Desarrollo Educativo **11**(22) (Jan 2021). https://doi.org/10.23913/ride.v11i22.848
15. Rodríguez Jara, M.A., Mena Lorca, A., Mena Lorca, J., Vásquez Saldias, P., Del Valle Leo, M.E.: Construcción cognitiva del conjunto solución de un sistema de ecuaciones lineales con dos incógnitas. Enseñanza de las Ciencias. Revista de investigación y experiencias didácticas **37**(1), 71–92 (Mar 2019). https://doi.org/10.5565/rev/ensciencias.2194
16. Serengil, S.I., Ozpinar, A.: Hyperextended lightface: A facial attribute analysis framework. In: 2021 International Conference on Engineering and Emerging Technologies (ICEET). p. 1–4. IEEE (Oct 2021). https://doi.org/10.1109/iceet53442.2021.9659697
17. Son, T.: Intelligent tutoring systems in mathematics education: A systematic literature review using the substitution, augmentation, modification, redefinition model. Computers **13**(10), 270 (Oct 2024). https://doi.org/10.3390/computers13100270
18. Tasoulas, T., Troussas, C., Mylonas, P., Sgouropoulou, C.: Affective computing in intelligent tutoring systems: Exploring insights and innovations. In: 2024 9th South-East Europe Design Automation, Computer Engineering, Computer Networks and Social Media Conference (SEEDA-CECNSM). p. 91–97. IEEE (Sep 2024). https://doi.org/10.1109/seeda-cecnsm63478.2024.00025
19. Teixeira, K.C.B.: Pedagogical strategies to enhance learning in a linear algebra course. PRIMUS **33**(2), 152–174 (Mar 2022). https://doi.org/10.1080/10511970.2022.2048928
20. Vazquez, L., Proctor, M.: Learning opportunities and challenges of sensor enabled intelligent tutoring systems on mobile computing devices. Journal of Advance Research in Computer Science & Engineering (ISSN: 2456-3552) **3**(2), 01–08 (Feb 2016). https://doi.org/10.53555/nncse.v3i2.431